LAB 3:

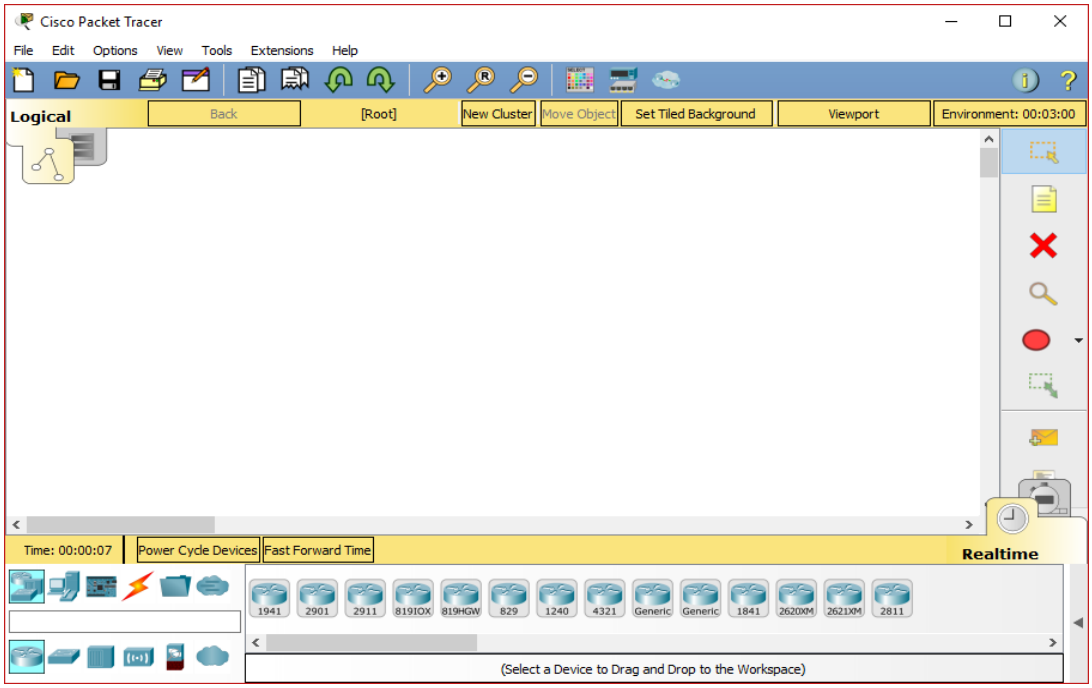
Simulation of network devices using cisco packet tracer

Objectives:

- To understand the basic concepts of computer networks through simulation.
- To study the working principles of network devices such as hub, switch, bridge, router, and repeater.
- To observe data transmission and packet flow between different network devices.

Theory:

Cisco Packet Tracer is a powerful network simulation platform used to design and test virtual network topologies. By providing a digital environment to configure devices like routers and switches, it eliminates the need for expensive physical hardware. It serves as an essential educational tool for mastering networking concepts, visualizing data flow, and honing troubleshooting skills.



Hub:

A **hub** is a foundational networking device that links multiple computers within a local network. Operating at the Physical Layer, it broadcasts incoming data packets to every connected port regardless of the intended destination. This "flooding" method often leads to **data collisions** and bandwidth congestion, making hubs significantly less efficient than modern switches.

Switch:

A switch is an improved version of a hub that connects multiple devices in a network. Unlike a hub, a switch sends data only to the intended device by using MAC addresses. This reduces unnecessary traffic and collisions in the network. Switches are commonly used in local area networks (LANs) because they provide better speed and performance.

Bridge:

A bridge is a network device used to divide a large network into smaller parts. It helps reduce network congestion by filtering traffic between different segments. A bridge checks the destination address before forwarding data, allowing only required data to pass through. This improves overall network efficiency.

Router:

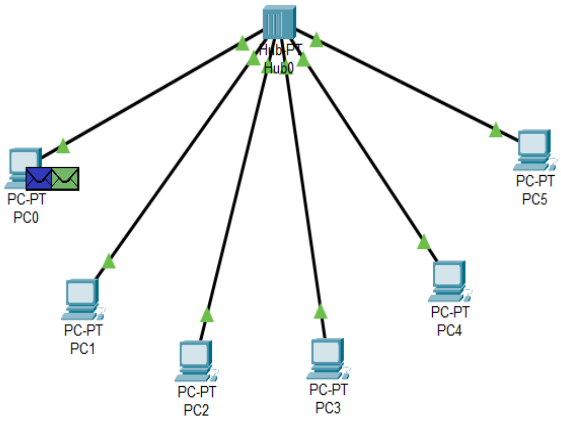
A router is a device that connects different networks together. It is mainly used to route data packets from one network to another using IP addresses. Routers choose the best path for data transmission and help in connecting local networks to the internet. They also provide better control and security in a network.

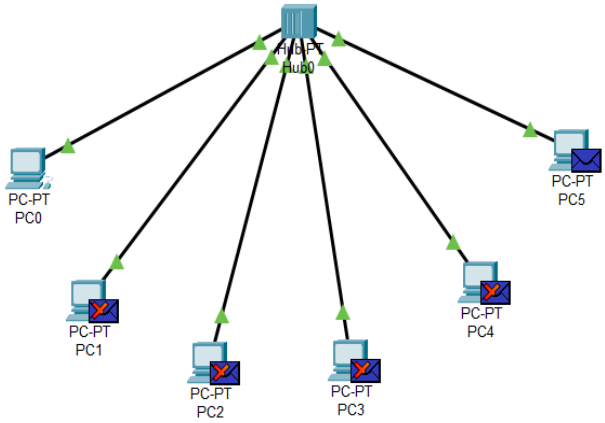
Repeater:

A repeater is used to extend the distance of a network. It works by receiving weak signals and regenerating them before sending them forward. Repeaters do not process or filter data; they only strengthen signals. They are useful when network cables need to cover long distances.

OBSERVATION:

Hub

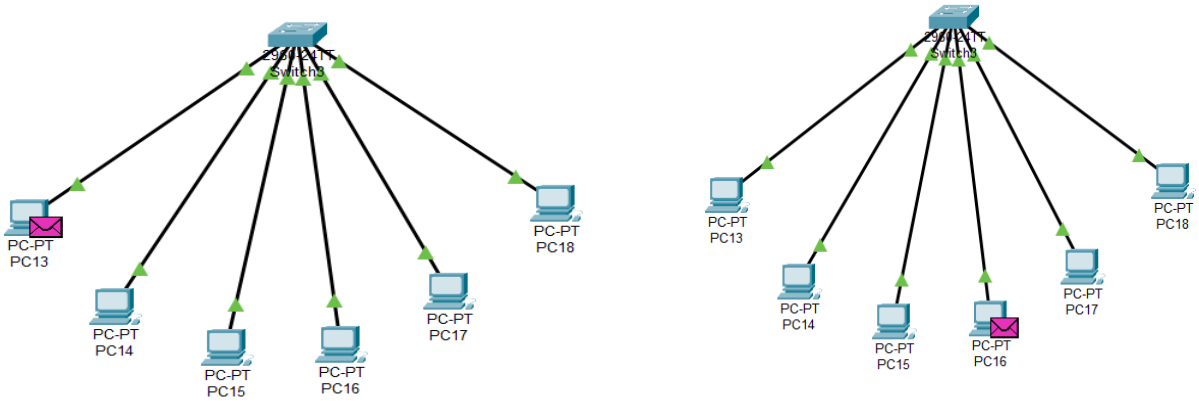




Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.001	PC0	Hub0	ICMP
	0.002	Hub0	PC1	ICMP
	0.002	Hub0	PC2	ICMP
	0.002	Hub0	PC3	ICMP
	0.002	Hub0	PC4	ICMP
	0.002	Hub0	PC5	ICMP

In this experiment, the hub was used to connect multiple devices in a single network. It was observed that when data was sent from one device, the hub broadcasted the data to all connected devices. This resulted in unnecessary traffic and chances of collision. The simulation helped in understanding why hubs are inefficient and are rarely used in modern networks.

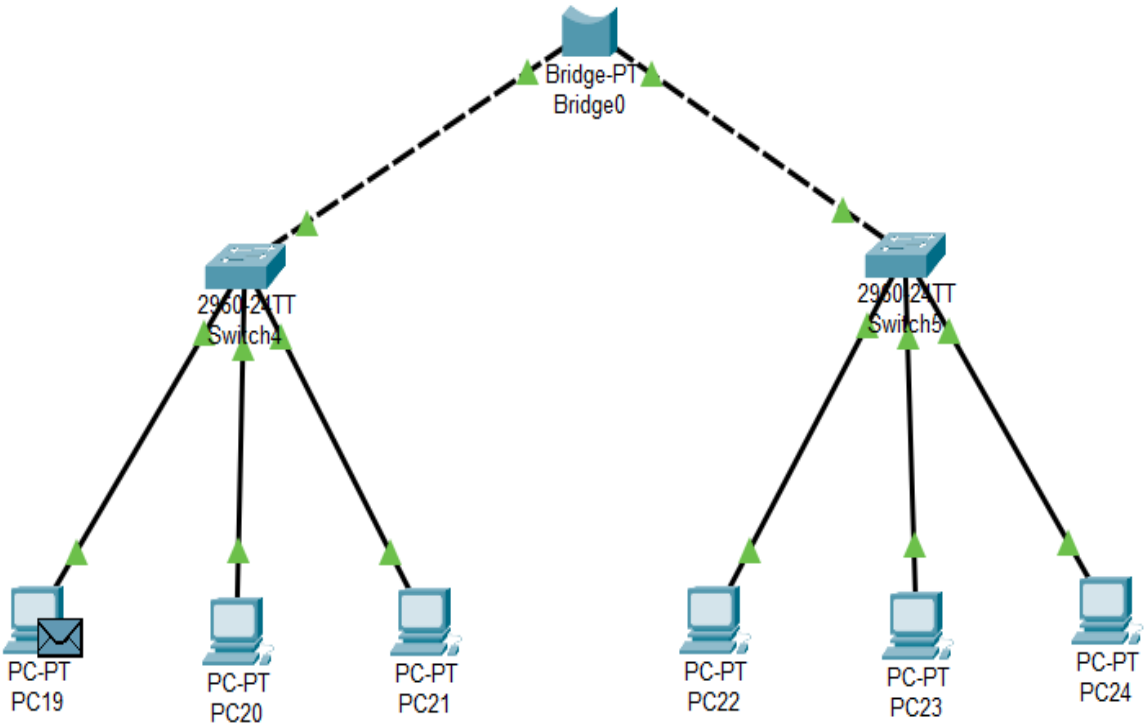
Switch



Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC13	ICMP
	0.001	PC13	Switch3	ICMP
	0.002	Switch3	PC16	ICMP

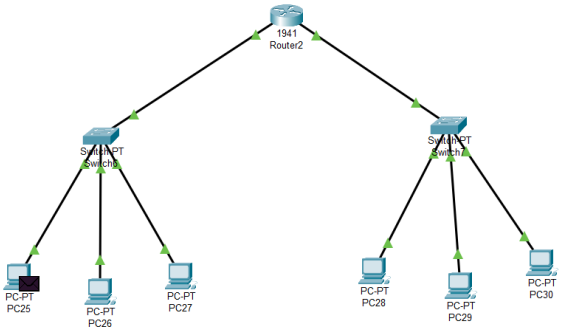
The switch showed better performance compared to the hub. During the simulation, data was forwarded only to the intended destination device instead of all devices. This reduced network traffic and collisions. The use of MAC addresses by the switch made communication faster and more reliable, showing why switches are commonly used in LANs.

Bridge

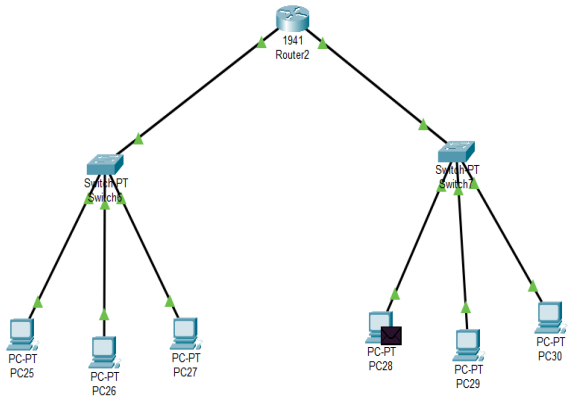


The bridge was used to divide the network into smaller segments. It was observed that the bridge filtered traffic and allowed only required data to pass between segments. This helped in reducing congestion in the network. The experiment demonstrated how bridges improve network efficiency by controlling data flow.

Router



Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC19	ICMP
	0.001	PC19	Switch4	ICMP
	0.002	Switch4	Bridge0	ICMP
	0.003	Bridge0	Switch5	ICMP
	0.004	Switch5	PC22	ICMP
	0.005	PC22	Switch5	ICMP

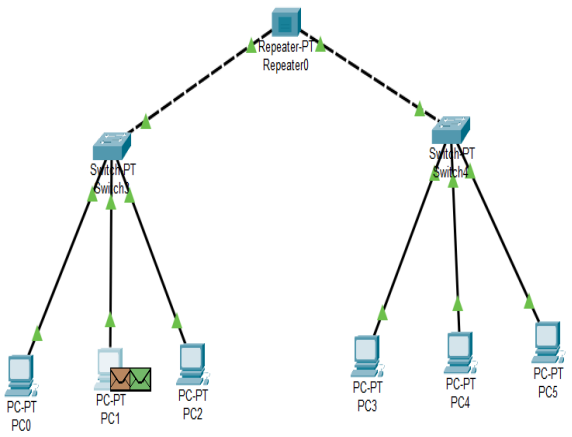


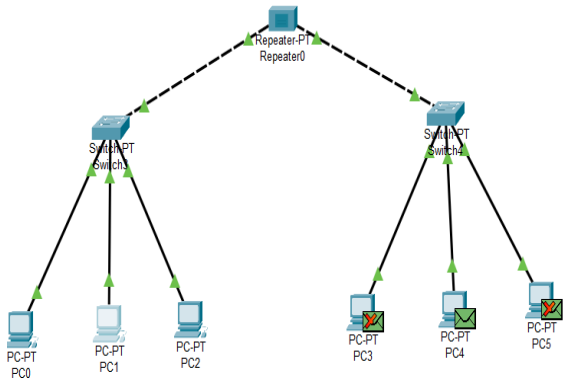
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC25	ICMP
	0.001	PC25	Switch6	ICMP
	0.002	Switch6	Router2	ICMP
	0.003	Router2	Switch7	ICMP
	0.004	Switch7	PC28	ICMP

```
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Request timed out.
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127
Reply from 10.10.10.2: bytes=32 time=8ms TTL=127

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 8ms, Average = 8ms
```

The router played an important role in connecting different networks. In the simulation, the router successfully routed data packets using IP addresses. It selected appropriate paths for data transmission, enabling communication between separate networks. This experiment helped in understanding how routers are essential for inter-network communication and internet access.

Repeater





Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC1	ICMP
	0.000	--	PC1	ARP
	0.001	PC1	Switch3	ARP
	0.002	Switch3	PC0	ARP
	0.002	Switch3	PC2	ARP
	0.002	Switch3	Repeater0	ARP
	0.003	Repeater0	Switch4	ARP
	0.004	Switch4	PC3	ARP
	0.004	Switch4	PC4	ARP
	0.004	Switch4	PC5	ARP

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=16ms TTL=128
Reply from 192.168.1.5: bytes=32 time=8ms TTL=128
Reply from 192.168.1.5: bytes=32 time=8ms TTL=128
Reply from 192.168.1.5: bytes=32 time=8ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 16ms, Average = 10ms
```

The repeater was used to extend the network distance. It was observed that weak signals were regenerated and forwarded, allowing data to travel longer distances without loss. The repeater did not filter or modify data, only strengthened the signal. This showed how repeaters help maintain signal quality in large networks.

Discussion and Conclusion:

In conclusion, this lab successfully utilized Cisco Packet Tracer to model and evaluate various networking components. By simulating the functions of hubs, switches, bridges,

routers, and repeaters, the exercise provided a risk-free environment to observe their practical operations. The results underscored how critical device selection is to meeting specific performance and architectural requirements. Ultimately, this session bridged the gap between theory and practice, strengthening core competencies in network design and configuration.